

CT024-3-3-DCOMS Distributed Computer Systems

Apply the concept of three-tier architecture, distributed systems and blockchain systems for solving issues in enterprise applications (C3, PL02)

Question 1

The ACE Bank, Ipoh, which has been in business for three generations and was founded in 1867, has its headquarters in Kuala Lumpur. More than 10,000 clients are served around Kuala Lumpur by over 70 personnel.

The ACE Bank IT department is searching for a server-based solution approach that can offer applications without requiring users to install them on their personal computers. They would like to have their administrators implement remote applications on a server in the company's data center or with a hosting service, and then deliver them to the users' desktops. The user can then access and use the app as though it were installed locally.

Which tier architecture helps implementing the said case? Justify your answer and discuss the proposed architecture in detail.

(25 marks)

Question 2

A large multinational company, Global Goods, Inc., specializes in producing consumer goods, including electronic devices. They are committed to ensuring ethical sourcing of raw materials and want to establish a transparent supply chain to maintain customer trust and comply with regulatory requirements. To achieve this, they decide to use leverage technology that is making rounds in recent times.

Which technology helps Global Goods to establish a transparent supply chain system? And discuss the components of the proposed technology.

(25 marks)